Slide 1: CASE 1 / 10

- Why Japan Recycles More Than Most Countries
- Product-specific laws, consumer-pays fees and traceability
- 84%
- Air-conditioner recycling rate (FY2005) vs 60% legal minimum
- 11.2 M
- Appliances collected in FY2004, up from 8.55 M in FY2001
- 63%
- Of 8.24 Mt plastic waste is burned for energy, only 25% recycled
- Key points
- Separate laws for appliances, packaging, small electronics
- Consumer pays at disposal; retailer or site collects; maker recycles
- Recycling tag lets people trace their appliance online
- Weak spot: heavy thermal recovery of plastics
- Lesson for India
- Copy take-back convenience, traceability and category-wise targets; avoid incineration dependence.

Slide 2: CASE 2 / 10

- Switzerland's E-Waste Success Story
- Advance recycling fee: pay first, return free, recycle at home
- 11.14 kg
- E-waste per person collected by SENS in 2024
- 1.4 Mt
- Collected since SENS began in 1990
- 16
- Accredited recyclers, all inside Switzerland
- Key points
- Producers pay a fee on every unit sold; SENS/Swico manage the fund
- Free return to any seller or 600+ collection points
- About 1,250 producer, importer and retailer partners
- Permit needed to export defective appliances
- Lesson for India
- Pre-fund recycling and make return free and easy; India's certificates lack an up-front fee.

Slide 3: CASE 3 / 10

- India's E-Waste (Management) Rules, 2022
- From voluntary compliance to certificate-based EPR
- 106
- EEE items covered (up from 21), in force 1 Apr 2023
- 60% to 80%
- Recycling target rising by 2027-28
- +151%
- E-waste growth 2017-18 to 2023-24 (17.78 lakh t)
- Key points
- All producers, recyclers and refurbishers register on CPCB portal
- Producers buy EPR certificates from registered recyclers
- Refurbishing certificates defer, not cancel, obligations
- Price floor about Rs 22/kg; penalties for non-compliance
- Lesson for India
- Strong design; success depends on audits and pulling the informal sector into the chain.

Slide 4: CASE 4 / 10

- India's EPR: Success or Paper Compliance?
- Real architecture, uneven verification
- 7,226
- Producers registered (Dec 2024)
- 22 lakh t
- Reported recycler capacity vs 17.8 lakh t generated
- ~90%
- E-waste still handled informally
- Key points
- Floor Rs 22/kg vs reported market bids as low as Rs 5.90/kg
- Delhi High Court challenge to the price band by major OEMs
- 600,000+ fake certificates found in plastic EPR audit (2023)
- CPCB capacity and budget seen as thin
- Lesson for India
- Tie certificates to physical evidence: weighbridge data, tracked transport, third-party audits.

Slide 5: CASE 5 / 10

- Bengaluru: India's E-Waste Capital?
- IT-driven waste, but formal recyclers get little of it
- 10-15%
- Share reaching authorised recyclers (KSPCB)
- ~1,000 t
- E-waste per month from IT companies alone
- 20,000-1 lakh t
- Annual estimates vary widely; no city measure
- Key points
- By volume Mumbai and Delhi rank above Bengaluru
- 27 authorised dismantlers and recyclers under 2011 Rules
- bE-Responsible pilot (2017): drop-offs at Bangalore One centres
- Consumers sell to kabadiwalas for quick cash
- Lesson for India
- Build city-level data, register kabadiwalas as collectors and enforce bulk-consumer rules.

Slide 6: CASE 6 / 10

- Mumbai's E-Waste Ecosystem
- Largest city generator, mostly informal chain
- #1
- City and Maharashtra #1 state for e-waste
- 83,086 t
- Maharashtra 2021-22 (37,099 t in 2020-21)
- 35%
- West India's share of national e-waste
- Key points
- Three tiers: generators, informal dealers, authorised recyclers
- Banks, corporates and IT firms are big bulk generators
- Licensed recycler example: 1,000 t capacity, 100-200 t processed
- Estimates conflict across years and methods
- Lesson for India
- Route bulk waste to authorised recyclers and give informal dealers a path to registration.

Slide 7: CASE 7 / 10

- Kerala's Clean Kerala Company
- State company + women collectors for non-biodegradable waste
- 86,296 t
- Collected by Haritha Karma Sena in 2025-26
- 37,481
- Women in 4,438 HKS units earning Rs 11.49 crore
- 5,286 km
- Roads built using shredded plastic
- Key points
- CKCL buys segregated waste from HKS and arranges disposal
- Local bodies run 1,200+ collection facilities
- Waste received grew from 7,657 t (2021-22) to 30,217 t (2022-23)
- Gap: few in-state recyclers, so recyclables go to Tamil Nadu
- Lesson for India
- Combine public aggregation with community workers; add e-waste and in-state recycling.

Slide 8: CASE 8 / 10

- Pune's SWaCH Model
- India's first wholly owned cooperative of waste pickers
- 10 lakh
- Households served daily (SWaCH 2025)
- 82,891 t
- Waste recycled annually
- Rs 113 cr
- Saved for Pune Municipal Corporation
- Key points
- Born from waste pickers' union KKPKP (1993); PMC agreement 2008
- Pairs of members serve 150-400 households each
- Income from user fees plus scrap sales
- Only about 43% of waste arrives segregated
- Lesson for India
- Recognise pickers as partners; make cooperatives the collection arm for EPR streams.

Slide 9: CASE 9 / 10

- Dharavi Recycling Ecosystem
- Mumbai's informal recycling engine and its uncertain future
- 5,000+
- Single-room recycling units
- 60-80%
- Of Mumbai's plastic, by estimates
- ~6,000 t
- Plastic a month in 13th Compound (one estimate)
- Key points
- Chain: pickers, kabadiwalas, sorters, crushers, washers, buyers
- Sorting into grades raises value from Rs 10-15/kg
- Workers earn about Rs 300-500 a day, with hazards
- Redevelopment threatens the whole network
- Lesson for India
- Measure, register and support it with EPR buy-back and safety, rather than replace it.

Slide 10: CASE 10 / 10

- Dell's Circular Economy Strategy
- Design, materials, packaging and take-back
- 2 bn lb
- Used electronics recovered since 2007 (Dell)
- 95%
- Leased devices returned and refurbished (India, 2021)
- 50%+
- 2030 goal for recycled or renewable product content
- Key points
- Goals: 1 t reused per 1 t sold; 100% recycled or renewable packaging
- Asset recovery services since 1996
- Recycled cobalt, aluminium, steel and ocean-bound plastic
- Data is company-reported; product content is the hard target
- Lesson for India
- Use bulk-consumer take-back and refurbishing; require verified circularity reporting.